

Project One Template

MAT350: Applied Linear Algebra

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July 13, 2026

Problem 1

Develop a system of linear equations for the network by writing an equation for each router (A, B, C, D, and E). Make sure to write your final answer as $A\mathbf{x}=\mathbf{b}$ where A is the 5×5 coefficient matrix, $\mathbf{x}$ is the 5×1 vector of unknowns, and $\mathbf{b}$ is a 5×1 vector of constants.

Solution:

Writing a conservation-of-flow equation (total inflow = total outflow) at each router gives:
Router A: $x_1 + 2x_2 = 100$
Router B: $x_1 + x_2 - x_3 - x_5 = 0$
Router C: $-x_2 + x_3 + x_5 = 50$ (i.e., $50 + x_2 = x_3 + x_5$)
Router D: $-x_2 + x_4 + x_5 = 120$ (i.e., $x_4 + x_5 = 120 + x_2$)
Router E: $x_2 + x_3 - x_4 + x_5 = 0$ (i.e., $x_2 + x_3 + x_5 = x_4$)
Writing this as $A\mathbf{x} = \mathbf{b}$:
 $A = [1 \ 2 \ 0 \ 0 \ 0; 1 \ 1 \ -1 \ 0 \ -1; 0 \ -1 \ 1 \ 0 \ 1; 0 \ -1 \ 0 \ 1 \ 1; 0 \ 1 \ 1 \ -1 \ 1]$
 $\mathbf{x} = [x_1; x_2; x_3; x_4; x_5]$
 $\mathbf{b} = [100; 0; 50; 120; 0]$

Problem 2

Use MATLAB to construct the augmented matrix $[A \ \mathbf{b}]$ and then perform row reduction using the `rref()` function. Write out your **reduced matrix and identify the free and basic variables of the system**.

Solution:

```
% Coefficient matrix A, constant vector b, and augmented matrix [A b]
A = [1 2 0 0 0; 1 1 -1 0 -1; 0 -1 1 0 1; 0 -1 0 1 1; 0 1 1 -1 1];
b = [100; 0; 50; 120; 0];
Ab = [A b]
```

```
Ab = 5x6
    1     2     0     0     0    100
    1     1    -1     0    -1     0
    0    -1     1     0     1     50
    0    -1     0     1     1    120
    0     1     1    -1     1     0
```

```
R = rref(Ab)
```

```
R = 5x6
    1     0     0     0     0     50
    0     1     0     0     0     25
    0     0     1     0     0     30
    0     0     0     1     0    100
    0     0     0     0     1     45
```

The reduced row echelon form places a pivot in every one of the first five columns (the identity matrix), so **x1, x2, x3, x4, and x5 are all basic variables and there are no free variables**. The system has exactly one solution: x1 = 50, x2 = 25, x3 = 30, x4 = 100, x5 = 45 (all in Mbps).

Problem 3

Use MATLAB to **compute the LU decomposition of A**, i.e., find $A = LU$. For this decomposition, find the transformed set of equations $Ly = b$, where $y = Ux$. Solve the system of equations $Ly = b$ for the unknown vector y .

Solution:

```
% LU factorization of A, then solve L*y = b for y
[L,U] = lu(A)
```

```
L = 5x5
    1.0000         0         0         0         0
    1.0000    1.0000         0         0         0
         0    1.0000    1.0000         0         0
         0    1.0000    0.5000    1.0000         0
         0   -1.0000         0   -1.0000    1.0000

U = 5x5
     1     2     0     0     0
     0    -1    -1     0    -1
     0     0     2     0     2
     0     0     0     1     1
     0     0     0     0     1
```

```
y = L\b
```

```
y = 5x1
    100
   -100
    150
    145
     45
```

Solving $Ly = b$ by forward substitution gives $y = [100; -100; 150; 145; 45]$. Recall that $y = Ux$, so this vector is the transformed right-hand side we will use in Problem 5 to recover x .

Problem 4

Use MATLAB to **compute the inverse** of U using the `inv()` function.

Solution:

```
% Inverse of U
Uinv = inv(U)
```

```
Uinv = 5x5
    1.0000    2.0000    1.0000         0         0
```

0	-1.0000	-0.5000	0	0
0	0	0.5000	0	-1.0000
0	0	0	1.0000	-1.0000
0	0	0	0	1.0000

Problem 5

Compute the solution to the original system of equations by transforming $\mathbf{y}$ into $\mathbf{x}$, i.e., compute $\mathbf{x} = \text{inv}(\mathbf{U})\mathbf{y}$.

Solution:

```
% Recover x from y via x = inv(U)*y
x = Uinv*y
```

```
x = 5x1
    50
    25
    30
   100
    45
```

Problem 6

Check your answer for x_1 using Cramer's Rule. Use MATLAB to compute the required determinants using the `det()` function.

Solution:

```
% Cramer's Rule for x1: replace column 1 of A with b
A1 = A;
A1(:,1) = b;
detA = det(A)
```

```
detA =
    -2
```

```
detA1 = det(A1)
```

```
detA1 =
   -100
```

```
x1 = detA1/detA
```

```
x1 =
    50
```

Cramer's Rule gives $x_1 = \det(A_1)/\det(A) = -100 / -2 = 50$, which matches the value of x_1 found in Problems 2 and 5, confirming the solution.

Problem 7

The Project One Table Template, provided in the Project One Supporting Materials section in Brightspace, shows the recommended throughput capacity of each link in the network. Put your solution for the system of equations in the third column so it can be easily compared to the maximum capacity in the second column. In the fourth column of the table, provide recommendations for how the network should be modified based on your network throughput analysis findings. The modification options can be No Change, Remove Link, or Upgrade Link. In the final column, explain how you arrived at your recommendation.

Solution:

Completed network analysis table (see also MAT_350_Project_One_Table_Completed.docx). Solution flows (Mbps): $x_1 = 50$, $x_2 = 25$, $x_3 = 30$, $x_4 = 100$, $x_5 = 45$. Recommendations: No Change for x_1 , x_2 , and x_3 ; Upgrade Link for x_4 (exactly at 100 Mbps capacity) and x_5 (45/50 Mbps, 90% utilization).

Network Link	Recommended Capacity (Mbps)	Solution	Recommendation	Explanation
x_1	60	50	No Change	The computed flow is 50 Mbps against a recommended capacity of 60 Mbps (about 83% utilization). The link is not exceeding capacity, so no modification is needed.
x_2	50	25	No Change	The computed flow is 25 Mbps against a recommended capacity of 50 Mbps (50% utilization). This link has ample headroom, so no change is required.
x_3	100	30	No Change	The computed flow is 30 Mbps against a recommended capacity of 100 Mbps (30% utilization). The link is well under capacity and does not need to be modified.
x_4	100	100	Upgrade Link	The computed flow is exactly 100 Mbps, matching its recommended capacity of 100 Mbps (100% utilization). With zero headroom, this link is at risk of reaching capacity if traffic increases, so it should be upgraded.
x_5	50	45	Upgrade Link	The computed flow is 45 Mbps against a recommended capacity of 50 Mbps (90% utilization). With only 5 Mbps of headroom remaining, this link is close to its limit and should be upgraded to reduce the risk of reaching capacity.

MAT 350 Project One Table Template

Complete this template by replacing the bracketed text with the relevant information.

Network Link	Recommended Capacity (Mbps)	Solution	Recommendation	Explanation
x_1	60	50	No Change	The computed flow is 50 Mbps against a recommended capacity of 60 Mbps (about 83% utilization). The link is not exceeding capacity, so no modification is needed.
x_2	50	25	No Change	The computed flow is 25 Mbps against a recommended capacity of 50 Mbps (50% utilization). This link has ample headroom, so no change is required.
x_3	100	30	No Change	The computed flow is 30 Mbps against a recommended capacity of 100 Mbps (30% utilization). The link is well under capacity and does not need to be modified.
x_4	100	100	Upgrade Link	The computed flow is exactly 100 Mbps, matching its recommended capacity of 100 Mbps (100% utilization). With zero headroom, this link is at risk of reaching capacity if traffic increases, so it should be upgraded.
x_5	50	45	Upgrade Link	The computed flow is 45 Mbps against a recommended capacity of 50 Mbps (90% utilization). With only 5 Mbps of headroom remaining, this link is close to its limit and should be upgraded to reduce the risk of reaching capacity.